A COCONE OF SCHEMES IS A COLIMIT IF IT IS FPQC LOCALLY A COLIMIT ON OVERLAPS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4182-4197 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 4199-4204, the complete author proof (4 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Descent theory's characterization of fpqc coverings as universal effective epimorphic families (Descent, fpqc-universal-effective-epimorphisms) and Categories' language of cocones; cited inputs with the mapping-property reduction written in the retained span.

Source reference. Original lemma 0G2U: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 4204–4204. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 023Q: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 4204–4204. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{I} \to (Sch/S)_{fppf}$, $i \mapsto X_i$ be a diagram of schemes. Let $(W, X_i \to W)$ be a cocone for the diagram in the category of schemes (Categories, Remark 0G2U). If there exists a fpqc covering $\{W_a \to W\}_{a \in A}$ of schemes such that*

1. *for all $a \in A$ we have $W_a = \operatorname{colim} X_i \times_W W_a$ in the category of schemes, and*
2. *for all $a, b \in A$ we have $W_a \times_W W_b = \operatorname{colim} X_i \times_W W_a \times_W W_b$ in the category of schemes,*

then $W = \operatorname{colim} X_i$ in the category of schemes.

Proof. Namely, for a scheme T a morphism $W \to T$ is the same thing as collection of morphism $W_a \to T$, $a \in A$ which agree on the overlaps $W_a \times_W W_b$, see Descent, Lemma 023Q. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.